

The seven-cycle case of the maximal anti-Ramsey conjecture

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Abstract

The maximal anti-Ramsey function $\chi_S(n, e, G)$ is the minimum number of colors needed to color the edges of some n -vertex, e -edge graph so that every copy of G is rainbow. Burr, Erdős, Graham and Sós conjectured that $\chi_S(n, \lfloor n^2/4 \rfloor + 1, C_{2k+1}) \sim n^2/8$ for every fixed $k \geq 3$. Bucić, Chen and Ma proved it for $k \geq 4$. We prove the remaining seven-cycle case, completing the proof of the conjecture. More generally, for $q = e/n^2$ we obtain

$$\chi_S(n, e, C_7) \geq \left(q - \frac{1}{8} + \frac{1}{4} \sqrt{q - \frac{1}{4}} \right) n^2 - o(n^2)$$

uniformly over $\lfloor n^2/4 \rfloor + 1 \leq e \leq \binom{n}{2}$. We also give a construction showing that the asymptotic formula for longer odd cycles fails for C_7 at every fixed density $1/4 < q < 1/2$.

1 Introduction

An edge-colored graph is *rainbow* if all its edges have different colors. Classical anti-Ramsey theory began with Erdős, Simonovits and Sós [5], who studied how many colors force a rainbow copy of a fixed graph in an edge-coloring of K_n .

For a fixed graph G , the *maximal anti-Ramsey function* $\chi_S(n, e, G)$, introduced and studied in [4, 3], is the minimum number of colors needed to color the edges of some n -vertex, e -edge graph so that every copy of G is rainbow. Deleting edges shows that the same minimum is obtained if graphs with at least e edges are allowed, as in [2].

For connected bipartite G that are not complete bipartite, Sárközy and Selkow [9, Theorem 4] proved that $\chi_S(n, e, G)/n \rightarrow \infty$ whenever e/n^2 stays bounded away from zero.

For odd cycles at $e = \lfloor n^2/4 \rfloor + 1$, the values of $\chi_S(n, e, C_3)$ and $\chi_S(n, e, C_5)$ are 3 and $\lfloor n/2 \rfloor + 3$, respectively, for sufficiently large n . The latter result is due to Erdős and Simonovits, as reported in [4, Sections 3–4]. For every fixed odd cycle of length at least seven, Burr, Erdős, Graham and Sós [4, Theorem 5.1] proved a quadratic lower bound.

They conjectured that $\chi_S(n, \lfloor n^2/4 \rfloor + 1, C_{2k+1}) \sim n^2/8$ for every fixed $k \geq 3$ [4, p. 270]. This is Erdős Problem 809 [1]. Bucić, Chen and Ma [2] proved the conjecture for $k \geq 4$. We prove it for $k = 3$.

Theorem 1.1. *As $n \rightarrow \infty$,*

$$\chi_S(n, \lfloor n^2/4 \rfloor + 1, C_7) = \frac{n^2}{8} + o(n^2).$$

The upper bound was observed in [4] (see also [2, Section 2]); it comes from the following construction. Take two disjoint cliques of nearly equal orders, color each injectively, and reuse the colors between the cliques. With a slight imbalance in their orders, one obtains the required number of edges. A modification using two cliques with one common vertex gives the sharper upper bound $n^2/8 + O(n)$. We first prove the matching lower bound of $n^2/8 - o(n^2)$ colors. We then use this threshold bound together with our weighted inequality to extend the lower bound to the full density range.

Theorem 1.2. *For $\lfloor n^2/4 \rfloor + 1 \leq e \leq \binom{n}{2}$, put $q = e/n^2$. Then, as $n \rightarrow \infty$,*

$$\chi_S(n, e, C_7) \geq \left(q - \frac{1}{8} + \frac{1}{4} \sqrt{q - \frac{1}{4}} \right) n^2 - o(n^2), \quad (1)$$

where the error term is uniform in e .

Bucić, Chen and Ma [2, Theorem 1.2] determine the exact asymptotic coefficient for C_{2k+1} , $k \geq 4$, throughout this range: it is $q/2 + \frac{1}{2}\sqrt{q-1/4}$. Writing $t = \sqrt{q-1/4}$, their coefficient exceeds the one in (1) by $t(1-2t)/4$, which is positive for $1/4 < q < 1/2$. Theorem 1.2 gives a lower bound; we do not determine the exact asymptotic value at interior densities. At every fixed interior density, however, their formula fails for C_7 .

Theorem 1.3. *Fix $1/4 < q < 1/2$, and let $s \in (0, 1/2)$ be the unique solution of*

$$q = \frac{1}{2} - 3s^2 + 4s^3.$$

Then, as $n \rightarrow \infty$,

$$\chi_S(n, \lfloor qn^2 \rfloor, C_7) \leq (q - s^3)n^2 + O(n),$$

and $q - s^3 < q/2 + \frac{1}{2}\sqrt{q-1/4}$.

For example, $q = 3/8$ corresponds to $s = 1/4$. At this density, Theorems 1.2 and 1.3 give the lower and upper coefficients $(4 + \sqrt{2})/16 \approx 0.338$ and $23/64 \approx 0.359$, while the coefficient of [2] for longer odd cycles is $(3 + 2\sqrt{2})/16 \approx 0.364$. We prove Theorem 1.3 in Section 7, using a chain of five cliques.

For the lower bounds, we first study closed walks of length seven. Two edges joined by suitable walks of lengths two and three lie on such a walk. This observation leads to a weighted inequality in Sections 2 and 3. A regularity argument then allows us to use it for cycles: after deleting at most ηn^2 edges, no closed seven-walk contains two distinct edges of the same color (Corollary 4.2). When q stays bounded away from $1/4$, the cleaned graph still has density above $1/4$, and uniform vertex weights give Theorem 1.2.

Near the Turán threshold, these deletions may remove the single extra edge, while the weighted inequality requires density strictly greater than $1/4$. We separate graphs according to the variance of their normalized degrees $d(v)/n$. When this variance stays bounded away from zero, shifting a little weight toward vertices of above-average degree restores the required density. When the variance tends to zero, we can remove $o(n)$ vertices and obtain minimum degree $n/2 - o(n)$ while preserving the edge excess. Section 5 gives a direct color bound for these nearly regular graphs.

The arguments in Sections 2–3, 4, and 5 are independent of one another. Section 6 combines the two variance cases to prove Theorem 1.1, then uses it to complete the density bound as q approaches $1/4$.

Bucić, Chen and Ma [2, Sections 2 and 4] use induction on n to reduce to graphs of large minimum degree, where they construct large families of edges any two of which lie on a common C_{2k+1} . This requires the two short paths joining a pair of such edges to have total length at most $2k - 1$, which for $k = 3$ fails in their denser case, where e/n^2 is bounded away from $1/4$ and the paths can need $3 + 4$ edges. At the end of their proof they remark that a more involved “stability” argument, which they do not present, handles their near-threshold case for $k = 3$, leaving the denser case as the main obstacle. Theorem 1.3 shows that this obstacle is genuine: their induction carries the bound of [2, Theorem 1.2] through the whole range of e , and for C_7 that bound fails at every fixed interior density. Our proof does not use induction on n ; apart from Section 5, which adapts short-path arguments from their near-threshold case to nearly regular graphs, it rests on the weighted inequality for closed seven-walks, fractional coloring, and regularity.

Formalization. Theorem 1.1 has been formalized in Lean 4 using Mathlib; the source code is available in [8].

Notation. All graphs are finite and simple. Copies of graphs need not be induced. An r -walk or r -path has r edges. A coloring is C_7 -rainbow if every copy of C_7 is rainbow. We normalize edge density as $q = e/n^2$.

2 A weighted bound for two- and three-walks

Suppose that edges ab and cd are joined by a two-edge walk from a to c and a three-edge walk from b to d . Together they form a closed seven-walk containing both edges. We therefore seek large sets of vertices in which every pair can be joined by walks of both lengths.

Let G be a graph, and let A be its adjacency matrix. Let $w = (w_i)_i$ be a nonnegative column vector of vertex weights. Put

$$W = \mathbf{1}^\top w, \quad d = Aw, \quad Q = \frac{1}{2} w^\top Aw,$$

where $\mathbf{1}$ is the all-ones vector. Thus d_i is the weighted degree of vertex i . For a vertex set K , write $w(K) = \sum_{i \in K} w_i$.

We introduce some terminology for the walk conditions that will recur in the proof. Note that $(A^r)_{ij} > 0$ if and only if an r -edge walk in G joins i to j . Call a vertex i *triangular* if $(A^3)_{ii} > 0$. A set K is a *joint clique* if $(A^2)_{ij} > 0$ and $(A^3)_{ij} > 0$ for every $i, j \in K$, including $i = j$. Thus every vertex of a joint clique is triangular. The next theorem shows that density above $W^2/4$ forces a joint clique containing more than half the total weight.

Theorem 2.1. *Let G and w be as above. If every joint clique in G has weight at most $s > 0$, then*

$$Q \leq \frac{s^2 + (W - s)^2}{2}. \quad (2)$$

Consequently, if $Q > W^2/4$, some joint clique has weight at least

$$\frac{W}{2} + \sqrt{Q - \frac{W^2}{4}}. \quad (3)$$

We use the following weighted version of Hajnal's clique intersection lemma [6]. To apply it to joint cliques, form an auxiliary graph on the triangular vertices, joining distinct i, j when both walk conditions hold. Its cliques are precisely the joint cliques.

Lemma 2.2 (Hajnal's lemma, weighted version). *In a finite graph with nonnegative vertex weights, if $C_1, \dots, C_r$ are maximum-weight cliques, each of weight s , then*

$$w\left(\bigcap_{j=1}^r C_j\right) + w\left(\bigcup_{j=1}^r C_j\right) \geq 2s.$$

Proof. Let I, U be the intersection and union of the first several cliques, and add a maximum clique C . The set $I \cup (C \cap U)$ is a clique, so $w(I \cup (C \cap U)) = w(I \setminus C) + w(C \cap U) \leq s = w(C)$. Thus $w(I \setminus C) \leq w(C \setminus U)$, and it follows that $w(I \cap C) + w(U \cup C) \geq w(I) + w(U)$. The assertion follows by induction, starting with one clique, for which the sum is $2s$. $\square$

Proof of Theorem 2.1. We prove (2) by optimizing over the vertex weights on G , with s fixed. The admissible weights form the closed polyhedron

$$\mathcal{P}_s = \{w \geq 0 : w(K) \leq s \text{ for every joint clique } K\}.$$

We will maximize the difference between the two sides of (2),

$$F_s(w) = Q - \frac{s^2 + (W - s)^2}{2} = -\frac{1}{2} w^\top (J - A) w + sW - s^2,$$

where J is the all-ones matrix.

Since $A_{ii} = 0$ for every i , dropping nonpositive off-diagonal terms gives

$$F_s(w) \leq -\frac{1}{2} \sum_i w_i^2 + s \sum_i w_i - s^2.$$

The right-hand side tends to $-\infty$ as $\|w\|_2 \rightarrow \infty$, so F_s attains a global maximum on $\mathcal{P}_s$. Suppose the maximum is positive, attained at $w \in \mathcal{P}_s$. Set $u = W - s$. Since $Q \leq W^2/2$, positivity implies $u > 0$.

Since A is symmetric,

$$\nabla F_s(w) = Aw - (W - s)\mathbf{1} = d - u\mathbf{1}.$$

Decreasing a positive coordinate preserves feasibility. Its first-order condition is therefore

$$d_i \geq u \quad (w_i > 0). \quad (4)$$

We claim that $d_i \leq s$ for every i . Otherwise take p with $d_p > s$. If $w_i > 0$ and $A_{pi} = 1$, then pi must be in a triangle of G . Indeed, if $N_G(p) \cap N_G(i)$ were empty, their weighted neighborhoods would be disjoint, so $d_i \leq W - d_p < u$, contrary to (4). Now all positive-weight neighbors of p form a joint clique: a, p, b is a two-walk between any two of them, and if c is the third vertex of a triangle containing pa , then a, c, p, b is a three-walk. This works also when $a = b$. Its weight is $d_p > s$, a contradiction. Hence

$$d_i \leq s \quad \text{for all } i. \quad (5)$$

In particular $2Q = w^\top d \leq sW$. Positivity of $F_s(w)$ now gives $0 < u < s$, and so $W < 2s$.

Since $\mathcal{P}_s$ is a polyhedron, the gradient at a maximizer is a nonnegative combination of the outward normals to its binding constraints. Thus there are multipliers $\mu_K \geq 0$ for the clique constraints and $\nu_i \geq 0$ for nonnegativity such that

$$d_i - u = \sum_{K \ni i} \mu_K - \nu_i, \quad \mu_K(w(K) - s) = 0, \quad \nu_i w_i = 0. \quad (6)$$

Put $L = \sum_K \mu_K$. Multiplying (6) by w_i and summing yields $2Q - uW = sL$. The strict inequality $F_s(w) > 0$ implies

$$sL > s^2 + u^2 - u(s + u) = s(s - u),$$

so $L > 0$. All cliques with $\mu_K > 0$ have weight s and are therefore maximum-weight joint cliques. Lemma 2.2 says that their intersection has weight at least $2s - W = s - u > 0$. Choose a positive-weight vertex p in that intersection. Equation (6) gives $d_p - u = L > s - u$, contradicting (5). This proves (2).

Let M be the maximum joint-clique weight when $Q > W^2/4$. Equation (2) with $s = W/2$ shows $M > W/2$. Apply it again with $s = M$ and solve the resulting quadratic inequality on the interval $M > W/2$ to obtain (3). $\square$

Remark 2.3. The bounds in Theorem 2.1 are asymptotically sharp. Fix $W > 0$ and $W/2 \leq M \leq W$. For each integer $t \geq 3$, take two disjoint copies of K_t , with vertex weights M/t in one and $(W - M)/t$ in the other. The largest joint clique has weight M , and

$$Q = \left(1 - \frac{1}{t}\right) \frac{M^2 + (W - M)^2}{2}.$$

As $t \rightarrow \infty$, this approaches the bound in (2) with $s = M$. If $M > W/2$, then $Q > W^2/4$ for all sufficiently large t , and the lower bound in (3) tends to M .

3 Fractional colorings and the seven-cycle inequality

We continue in the setting of Section 2. Assume that $\sum_i w_i = 1$. To each edge ij of G , assign the weight $w_i w_j$. Then

$$Q = \sum_{ij \in E(G)} w_i w_j.$$

Define a *conflict graph* $J_{23} = J_{23}(G)$ with vertex set $E(G)$ by joining two edges if they admit orientations (a, b) and (c, d) for which

$$(A^2)_{ac} > 0 \quad \text{and} \quad (A^3)_{bd} > 0. \quad (7)$$

The two walks implied by (7), together with ab and cd , form a closed seven-walk. Consequently, if no closed seven-walk contains two distinct edges of the same color, each color class is an independent set of J_{23} . Corollary 4.2 shows how to achieve this condition by deleting a small number of edges when every C_7 is rainbow.

To account for these edge weights, we use fractional colorings. For a finite simple graph Γ , let $\mathcal{I}(\Gamma)$ be its nonempty independent sets; vertices in one such set may share a color. Given nonnegative numbers d_i specifying the required coverage of each vertex, define the fractional coloring cost by

$$\Phi(\Gamma; d) = \min \left\{ \sum_{I \in \mathcal{I}(\Gamma)} z_I : z_I \geq 0, \quad \sum_{I \ni i} z_I \geq d_i \quad (\forall i) \right\}. \quad (8)$$

Finite linear programming duality gives

$$\Phi(\Gamma; d) = \max \left\{ \sum_i d_i y_i : y_i \geq 0, \sum_{i \in I} y_i \leq 1 \ (\forall I \in \mathcal{I}(\Gamma)) \right\}. \quad (9)$$

Call an allocation z *exact* if $\sum_{I \ni i} z_I = d_i$ for every i . Any optimum in (8) can be made exact: when a vertex receives more than d_i , split weights z_I as needed and remove the vertex from some of the corresponding independent sets. Discard empty sets; the total cost does not increase.

Using only singleton independent sets costs $\sum_i d_i$. The next lemma bounds the saving from using larger independent sets, with a stronger bound when a clique contains at least half the vertex weight.

Lemma 3.1 (Fractional coloring savings). *Let $v_i \geq 0$ sum to one and let $\alpha_i \geq 0$ be feasible in (9). With $R = \Phi(\Gamma; (v_i \alpha_i)_i)$,*

$$\sum_i v_i \alpha_i - R \leq \frac{1}{4}. \quad (10)$$

If Γ contains a clique of v -mass $p \geq 1/2$, then

$$\sum_i v_i \alpha_i - R \leq p(1 - p). \quad (11)$$

Proof. The dual vector $y_i = \alpha_i$ gives $R \geq \sum_i v_i \alpha_i^2$. Singleton sets give $0 \leq \alpha_i \leq 1$, and $a(1 - a) \leq 1/4$ proves (10).

For (11), let L be the clique and take an exact optimal allocation z_I . Exact coverage expresses the saving as $\sum_I (|I| - 1) z_I$. We seek coefficients c_i such that $\sum_{i \in I} c_i \geq |I| - 1$ whenever $z_I > 0$; multiplying by z_I and summing will then bound the saving by $\sum_{i: \alpha_i > 0} c_i v_i \alpha_i$. If $z_I > 0$, exact coverage gives $v_i \alpha_i > 0$ for every $i \in I$, so $\alpha_i > 0$. For these vertices, set

$$c_i = \begin{cases} (1 - p)^2 / \alpha_i, & i \in L, \\ p^2 / \alpha_i, & i \notin L. \end{cases}$$

Every independent set I with $z_I > 0$ satisfies $\sum_{i \in I} c_i \geq |I| - 1$. Indeed, it contains at most one vertex of L . If it contains one with value a and $k \geq 1$ others with values b_j , Cauchy-Schwarz and dual feasibility give

$$\frac{(1 - p)^2}{a} + p^2 \sum_{j=1}^k \frac{1}{b_j} \geq \frac{(1 - p + pk)^2}{a + \sum_j b_j} \geq (1 - p + pk)^2 \geq k.$$

The last step uses $p \geq 1/2$ and $k \geq 1$; the case $k = 0$ is immediate. If I avoids L and has $k \geq 1$ vertices, then $p^2 \sum_{i \in I} 1/\alpha_i \geq p^2 k^2 \geq k - 1$. Exact coverage therefore yields

$$\begin{aligned} \sum_i v_i \alpha_i - R &= \sum_I (|I| - 1) z_I \leq \sum_{i: \alpha_i > 0} c_i v_i \alpha_i \\ &\leq (1 - p)^2 p + p^2 (1 - p) = p(1 - p). \end{aligned}$$

□

We apply this estimate to the conflict graph, with the edge weights $w_i w_j$ as the required coverages. If no closed seven-walk repeats a color on distinct edges, each color class is independent in J_{23} . With uniform vertex weights $1/n$, assigning each nonempty such class weight $1/n^2$ therefore shows that the fractional coloring cost C below is at most r/n^2 , where r is the number of colors.

Theorem 3.2. *Let G be a graph with nonnegative vertex weights w summing to 1, and put*

$$Q = \sum_{ij \in E(G)} w_i w_j, \quad C = \Phi(J_{23}(G); (w_i w_j)_{ij \in E(G)}).$$

If $Q > 1/4$, then

$$C \geq Q - \frac{1}{8} + \frac{1}{4} \sqrt{Q - \frac{1}{4}} > \frac{1}{8}. \quad (12)$$

Proof. Choose a joint clique K of mass $m = w(K) \geq \frac{1}{2} + \sqrt{Q - 1/4}$ by Theorem 2.1. Write $U = V(G) \setminus K$, $u = w(U) = 1 - m$, and let Q_K, Q_{KU}, Q_U be the edge masses inside K , across (K, U) , and inside U , respectively. Thus $Q = Q_K + Q_{KU} + Q_U$. We will show that $Q - C \leq u/4$; the lower bound on m will then give (12).

Edges within K form a clique in J_{23} , and each conflicts with every edge crossing from K to U . For the latter assertion, the two-walk joins their K endpoints, while a two-walk between vertices of K followed by the crossing edge gives the three-walk required in (7). Consequently, if C_{cut} denotes the fractional coloring cost of the crossing edges, duality gives $C \geq Q_K + C_{\text{cut}}$. If $u = 0$, this says $C = Q$; suppose $u > 0$.

Define a graph Γ on U by joining distinct x, y whenever a two- or a three-walk joins them in G . Put

$$g_x = \sum_{i \in K} w_i A_{ix}, \quad \rho_x = w_x g_x, \quad Q_{KU} = \sum_{x \in U} \rho_x.$$

An independent set of crossing edges in J_{23} has distinct outside endpoints, and these endpoints form an independent set in Γ . Two edges with the same outside endpoint conflict by a closed two-walk there and a three-walk between their K endpoints. If two different outside endpoints admit a two-walk, use the three-walk between K endpoints; if they admit a three-walk, use the two-walk between K endpoints. Project an exact allocation for the crossing edges to their outside endpoints. It covers ρ_x exactly, whence, with $R = \Phi(\Gamma; \rho)$,

$$C \geq Q_K + R, \quad Q - C \leq Q_U + Q_{KU} - R. \quad (13)$$

For any independent set I of Γ , the neighborhoods $N_G(x) \cap K$, $x \in I$, are pairwise disjoint: a common neighbor would give a two-walk. Hence $\sum_{x \in I} g_x \leq m$. Set $v_x = w_x/u$ and $\alpha_x = g_x/m$. The vector α is dual feasible, and $\rho_x = mu v_x \alpha_x$. Lemma 3.1 and homogeneity give

$$Q_{KU} - R \leq \frac{mu}{4}. \quad (14)$$

If Γ has a clique L of original weight $\ell \geq u/2$, the stronger conclusion is

$$Q_{KU} - R \leq \frac{m}{u} \ell(u - \ell). \quad (15)$$

To finish, compare the edge mass Q_U inside U with $u^2/4$. If $Q_U \leq u^2/4$, equations (13) and (14) give $Q - C \leq Q_U + mu/4 \leq (u^2 + mu)/4 = u/4$.

If $Q_U > u^2/4$, Theorem 2.1 yields a joint clique within U , allowing us to use the stronger savings bound. Its weight satisfies $\ell \geq u/2 + \sqrt{Q_U - u^2/4}$. Its walks in $G[U]$ are also walks in G , so it is a clique of Γ , and

$$\ell(u - \ell) \leq \frac{u^2}{2} - Q_U.$$

Since $m \geq u$, equations (13) and (15) imply

$$Q - C \leq Q_U + \frac{m}{u} \ell(u - \ell) \leq \frac{mu}{2} + \left(1 - \frac{m}{u}\right) Q_U \leq \frac{mu + u^2}{4} = \frac{u}{4}.$$

Thus always $C \geq Q - (1 - m)/4$. Inserting the lower bound on m proves (12). $\square$

4 From closed walks to cycles

We use Szemerédi's regularity lemma [10] and the embedding lemma of Komlós and Simonovits [7, Theorem 2.1] to pass from closed walks to cycles. The point needed here is that two specified edges can be kept fixed.

Lemma 4.1 (Seven-walk cleaning). *For every $\eta > 0$, if n is sufficiently large, then every n -vertex simple graph G has a spanning subgraph $H \subseteq G$, obtained by deleting at most ηn^2 edges, such that any two distinct edges belonging to a common closed seven-walk in H belong to a common C_7 in G .*

Proof. Choose $0 < d < 1$ small in terms of η , then a large integer t_0 , and finally $\varepsilon > 0$ small in terms of d and η . Apply the equitable regularity lemma to partition $V(G)$ into an exceptional set of size at most εn and $t \geq t_0$ equal-sized parts, each of size M . At most εt^2 pairs of parts are irregular. Delete edges meeting the exceptional set, edges within parts, edges in irregular pairs, and edges in pairs of density below d .

We also trim vertices with unusually small degree in a retained pair. In a pair of original density p , delete the edges of that pair incident to any vertex with fewer than $(p - \varepsilon)M$ neighbors in the opposite part. Regularity bounds the number of such vertices by εM on each side. The total deletion cost is $O(d + \varepsilon + t_0^{-1})n^2 < \eta n^2$. Denote the remaining graph by H .

Claim. *A walk $x_0, \dots, x_r$ in H with distinct endpoints and $3 \leq r \leq 5$ can be replaced by a path in G with the same endpoints, length and sequence of parts. The path can also avoid any set of at most three vertices other than its endpoints.*

Proof of the claim. Let V_j be the part containing x_j . Set $S_1 = N_G(x_0) \cap V_1$, $S_{r-1} = N_G(x_r) \cap V_{r-1}$, and $S_j = V_j$ for $2 \leq j \leq r-2$. Trimming gives $|S_j| \geq (d - \varepsilon)M$ for every j . After removing x_0, x_r and the forbidden vertices, each set still has at least $dM/2$ vertices for large M . Since there are at most four internal positions, we can greedily choose pairwise disjoint subsets $T_j \subseteq S_j$, each of size $\lfloor dM/8 \rfloor$. The slicing lemma [7, Fact 1.5] shows that consecutive pairs (T_j, T_{j+1}) are $O(\varepsilon/d)$ -regular with density at least $d - \varepsilon$. Since ε is sufficiently small, the embedding lemma [7, Theorem 2.1], applied to the path on $r-1$ vertices, gives a path $y_1, \dots, y_{r-1}$ with $y_j \in T_j$. Then $x_0, y_1, \dots, y_{r-1}, x_r$ is the required path. $\square$

We apply the claim to two distinct marked edges e, f in a closed seven-walk in H . Choose one occurrence of each edge. First suppose $e = ab$ and $f = cd$ are disjoint, with the chosen occurrences traversed as $a \rightarrow b$ and $c \rightarrow d$. Let P, Q be the intervening walks from b to c and from d to a , of lengths p, q . Since $p, q \geq 1$ and $p + q = 5$, interchanging the marked edges if necessary gives $(p, q) = (1, 4)$ or $(2, 3)$.

If $p = 1$, retain the path a, b, c, d and replace Q by a four-edge path avoiding b, c .

If $p = 2$, write $P = b, x, c$. When $x \notin \{a, d\}$, retain the path a, b, x, c, d and replace Q by a three-edge path avoiding b, x, c . If $x = a$, retain b, a, c, d and replace the four-walk formed by Q followed by ab ; if $x = d$, retain a, b, d, c and replace the four-walk formed by cd followed by Q . In these last two cases the new path avoids the two internal vertices of the retained path. Each choice gives a seven-cycle containing e, f .

Now suppose the marked edges share an endpoint, and write them ab, ac . It suffices to find an odd walk between b and c of length at most five. Backtracks extend it to length five, and the claim gives a five-edge path avoiding a . Together with ab, ac , this path forms the required seven-cycle.

Orient the closed walk so that the chosen occurrence of ab is traversed $b \rightarrow a$. If the chosen occurrence of ac is traversed $a \rightarrow c$, the complementary gaps run from a to a and from c to b ; let their lengths be p, q , with $p + q = 5$. If q is odd, use the c - b gap. If q is even, then $q \geq 2$, so ba , the a - a gap, and ac give an odd walk of length $p + 2 \leq 5$.

In the other orientation, the complementary gaps run from a to c and from a to b , and their lengths sum to five. Exactly one has even length. Prepend ba to the a - c gap if it has even length, or ca to the a - b gap otherwise. The resulting odd walk has length at most five, since the unused gap has length at least one. $\square$

We will use the following coloring consequence.

Corollary 4.2. *For every $\eta > 0$, if n is sufficiently large, then every n -vertex edge-colored simple graph G in which every C_7 is rainbow has a spanning subgraph $H \subseteq G$, obtained by deleting at most ηn^2 edges, such that no closed seven-walk in H contains two distinct edges of the same color.*

Proof. Take H from Lemma 4.1: two distinct edges of the same color in a closed seven-walk in H would lie together on a C_7 in G , contradicting the rainbow hypothesis. $\square$

5 Nearly regular graphs

We next prove the color bound when the normalized degree variance tends to zero. Lemma 5.3 will reduce this case to graphs of minimum degree $n/2 - o(n)$ by removing $o(n)$ vertices while preserving

the edge excess. In such graphs we find a large family of edges any two of which lie on a common seven-cycle. Call such a family C_7 -compatible. Its edges must all have different colors in a coloring where every seven-cycle is rainbow.

Lemma 5.1 (Nearly bipartite graphs). *Suppose $e(G_n) > \lfloor n^2/4 \rfloor$ and G_n can be made bipartite by deleting $o(n^2)$ edges. Then G_n has a C_7 -compatible family of $(1/8 - o(1))n^2$ edges.*

Proof. Take a maximum cut (A, B) . Write I for its internal edge count and M for its missing cross-edge count. Since

$$e(G) = |A||B| - M + I > \lfloor n^2/4 \rfloor \geq |A||B|,$$

we have $I > M$, while $I, M = o(n^2)$ and $|A|, |B| = n/2 + o(n)$. For a vertex v , let $s(v)$ be its internal degree, $d_\times(v)$ its cross degree, and $m(v)$ its missing cross degree. Maximality of the cut gives $s(v) \leq d_\times(v)$.

Fix $0 < \varepsilon < 1/16$, put $\kappa = \varepsilon/4$, and let

$$X = \{v : m(v) > \kappa n\}, \quad \rho = |X|/n = o(1), \quad \alpha = (1/4 - \varepsilon)n.$$

We claim that some internal edge has at least α common neighbors across the cut. Suppose otherwise. We will find a set $Z \subseteq X$ meeting every internal edge and satisfying $m(z) - s(z) \geq \varepsilon n$ for every $z \in Z$. This surplus, together with $|X| = o(n)$, will force $I \leq M$, a contradiction. Define

$$L = \{v : d_\times(v) < \alpha + \kappa n\}, \quad Z = L \cup \{v \notin L : m(v) \geq n/8\}.$$

For large n , $L \subseteq X$. If an internal edge vw has $w \notin X$, its common cross neighborhood has size at least $d_\times(v) - \kappa n$, so $v \in L$. Consequently, a vertex outside L has internal degree at most ρn . If an internal edge has neither endpoint in L , its endpoints have missing cross degrees summing to more than $n/4$. Thus $Z \subseteq X$ meets every internal edge.

For $z \in L$, using $s(z) \leq d_\times(z)$ and the balance of the cut gives

$$m(z) - s(z) > (3\varepsilon/2 - o(1))n.$$

For $z \in Z \setminus L$, one has $m(z) \geq n/8$ and $s(z) \leq \rho n$. Hence, for sufficiently large n , $m(z) - s(z) \geq \varepsilon n$ for all $z \in Z$. Since Z meets every internal edge, counting a missing cross edge twice only when both endpoints lie in Z gives

$$I \leq \sum_{z \in Z} s(z) \leq M + |Z \cap A||Z \cap B| - \varepsilon n|Z| \leq M + (\rho/4 - \varepsilon)n|Z| \leq M,$$

a contradiction. This proves the claim.

Let uv be the claimed internal edge, say in A , and set

$$S = (N(u) \cap N(v) \cap B) \setminus X, \quad A' = A \setminus (X \cup \{u, v\}).$$

Then $|S| \geq (1/4 - \varepsilon - o(1))n$, $|A'| = n/2 - o(n)$, and every vertex outside X misses at most κn cross neighbors. Thus $G[A', S]$ has at least $(1/8 - O(\varepsilon) - o(1))n^2$ edges.

Any two of these edges lie on a common seven-cycle. For disjoint edges ab, cd , with $a, c \in A'$ and $b, d \in S$, choose a common cross neighbor $t \in B \setminus \{b, d\}$ of a, c ; then

$$u, v, b, a, t, c, d, u$$

is a seven-cycle. For edges ab, ad sharing $a \in A'$, choose $s \in S \setminus \{b, d\}$ and then a common neighbor $c \in A'$ of s, b , avoiding a ; use u, s, c, b, a, d, v, u . For edges ab, cb sharing $b \in S$, choose distinct $s, t \in S \setminus \{b\}$ with $as, ct \in E(G)$; use u, s, a, b, c, t, v, u . The candidate sets in each case have linear size, so the displayed vertices can be chosen distinct. Let $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$. $\square$

The next proof uses a short-path argument similar to [2, Claim 2 in Section 4]. Since we require paths of exactly three edges, the obstruction can also be a nearly bipartite graph; Lemma 5.1 handles this case.

Lemma 5.2 (Nearly regular graphs). *Suppose*

$$e(G_n) > \lfloor n^2/4 \rfloor, \quad e(G_n) = n^2/4 + o(n^2), \quad \delta(G_n) \geq n/2 - o(n).$$

Then every C_7 -rainbow coloring of G_n uses at least $n^2/8 - o(n^2)$ colors.

Proof. Suppose first that every pair of distinct vertices can be joined by a three-edge path avoiding any prescribed set of at most three other vertices. Choose any vertex p , and mark every edge incident to $N(p)$ except the edges incident to p . Two disjoint marked edges can be written xy, zw with $x, z \in N(p)$. The four-edge path y, x, p, z, w contains both; close it by a three-path from w to y avoiding its three internal vertices. For two adjacent marked edges, greedily extend their two-edge path to a four-edge path and close it the same way. The minimum-degree hypothesis yields the two new vertices. Hence the marked edges are C_7 -compatible. Counting degrees at $N(p)$, with each edge counted at most twice, bounds their number below by

$$\frac{|N(p)|\delta(G)}{2} - O(n) \geq \frac{\delta(G)^2}{2} - O(n) \geq \frac{n^2}{8} - o(n^2).$$

Otherwise choose distinct x, y and a set F of at most three vertices such that no three-path from x to y avoids F . Put $A = N(x) \setminus (F \cup \{y\})$ and $B = N(y) \setminus (F \cup \{x\})$. Both sets have size $n/2 - o(n)$, and no edge joins A to B . If $A \cap B \neq \emptyset$, a vertex in the intersection has no neighbors in $A \cup B$. The degree bound forces $|A \cup B| \leq n/2 + o(n)$, so A and B differ by only $o(n)$ vertices and $e(G[A]) = o(n^2)$. Summing degrees over A shows that the cut $(A, V \setminus A)$ has $n^2/4 - o(n^2)$ edges. Only $o(n^2)$ edges are internal to it, so Lemma 5.1 applies.

If $A \cap B = \emptyset$, the two sets are almost-half-sized and cover all but $o(n)$ vertices. Every vertex of A has $|A| - o(n)$ neighbors inside A , because it has none in B . Thus $G[A]$ has $n^2/8 - o(n^2)$ edges. Any two of its edges lie on a common seven-cycle: pairs have $|A| - o(n)$ common neighbors, and any two vertices have a three-path avoiding a fixed bounded set. Use disjoint two- and three-path connectors for disjoint edges; for adjacent edges extend their two-edge path to length four and close with a three-path. All edges of $G[A]$ have different colors. $\square$

Lemma 5.3 (Vanishing variance). *Suppose $e(G_n) > \lfloor n^2/4 \rfloor$, $e(G_n) = n^2/4 + o(n^2)$, and*

$$\sigma_n^2 = \frac{1}{n} \sum_{v \in V(G_n)} \left(\frac{d(v)}{n} - \frac{2e(G_n)}{n^2} \right)^2 \rightarrow 0.$$

Then every C_7 -rainbow coloring uses at least $n^2/8 - o(n^2)$ colors.

Proof. Choose $a_n \rightarrow 0$ with $a_n n \rightarrow \infty$ and $\sigma_n^2 = o(a_n^3)$. Repeatedly delete a vertex of current degree less than $N/2 - a_n n$, where N is the current order. Every deletion preserves strict excess over $\lfloor N^2/4 \rfloor$, since $\lfloor N^2/4 \rfloor - \lfloor (N-1)^2/4 \rfloor = \lfloor N/2 \rfloor$ and, for large n , the removed degree is smaller.

The small variance prevents this process from deleting many vertices. Before $a_n n$ deletions, each deleted vertex had original degree at most $n/2 - a_n n/2$. Since the original mean degree is at least $n/2$, Chebyshev's inequality bounds the number of such vertices by $4\sigma_n^2 n/a_n^2 = o(a_n n)$. Thus the process stops after $o(n)$ deletions. The remaining graph has $N = n - o(n)$ vertices, $e = N^2/4 + o(n^2) > \lfloor N^2/4 \rfloor$, and minimum degree at least $N/2 - o(n)$. Lemma 5.2 gives $N^2/8 - o(N^2) = n^2/8 - o(n^2)$ colors on the remaining edges. $\square$

6 Proofs of the main theorems

Lower bound in Theorem 1.1. Suppose the lower bound fails. Then for some $0 < \gamma < 1/8$ there is an unbounded sequence of graphs G_n with every C_7 rainbow and

$$e(G_n) = \lfloor n^2/4 \rfloor + 1, \quad r(G_n) \leq (1/8 - \gamma)n^2, \quad (16)$$

where $r(G_n)$ is the number of colors used. Pass to a subsequence on which the normalized degree variance σ_n^2 converges. Lemma 5.3 rules out the case that the limit is zero. Thus there is some $v > 0$ such that $\sigma_n^2 \geq v$ after passing to a further subsequence.

Apply Corollary 4.2 with a fixed $\eta > 0$, chosen small in terms of γv . Let $H \subseteq G_n$ be the cleaned graph, with adjacency matrix A_H , and set

$$D_i = \frac{d_H(i)}{n}, \quad q_H = \frac{e(H)}{n^2}, \quad \sigma_H^2 = \frac{1}{n} \sum_i (D_i - 2q_H)^2.$$

Let $D_i^G = d_{G_n}(i)/n$. Since H is obtained by deleting at most ηn^2 edges,

$$\frac{1}{n} \sum_i (D_i^G - D_i) = \frac{2(e(G_n) - e(H))}{n^2} \leq 2\eta.$$

Since $D_i, D_i^G \in [0, 1]$, expanding the variances gives

$$|\sigma_n^2 - \sigma_H^2| \leq \frac{4}{n} \sum_i (D_i^G - D_i) \leq 8\eta.$$

Thus $\sigma_H^2 \geq v/2$ for sufficiently small η . Also $q_H \geq 1/4 - \eta$.

We restore the density lost in cleaning by shifting weight toward vertices of above-average degree. Put

$$z_i = \frac{D_i - 2q_H}{n}, \quad w_i = \frac{1}{n} + \gamma z_i.$$

These weights are positive, sum to one, and satisfy $w_i \leq (1 + \gamma)/n$. The linear increase in weighted edge density is exactly $\gamma\sigma_H^2$. Cauchy-Schwarz gives $\|z\|_1^2 \leq \sigma_H^2$, which also bounds the quadratic error. Taking $\eta < (\gamma - \gamma^2/2)v/2$, we obtain

$$\begin{aligned} Q(w) &= \frac{1}{2} w^\top A_H w = q_H + \gamma\sigma_H^2 + \frac{\gamma^2}{2} z^\top A_H z \\ &\geq q_H + (\gamma - \gamma^2/2)\sigma_H^2 > \frac{1}{4}. \end{aligned} \tag{17}$$

By Corollary 4.2, each color class restricted to $E(H)$ is independent in $J_{23}(H)$. Give each nonempty such class c weight $\max_{ij \in c} w_i w_j$. The resulting allocation covers the weight of every edge of H . Theorem 3.2, applied to H with the weights w , and (16) imply

$$\frac{1}{8} < C \leq \sum_{\text{nonempty classes } c} \max_{ij \in c} w_i w_j \leq \frac{(1 + \gamma)^2}{n^2} r(G_n) \leq (1 + \gamma)^2 (1/8 - \gamma) < \frac{1}{8}. \quad \square$$

Proof of Theorem 1.2. Write

$$f(q) = q - \frac{1}{8} + \frac{1}{4} \sqrt{q - \frac{1}{4}} \quad (1/4 \leq q \leq 1/2).$$

Choose $\eta_n \downarrow 0$ slowly enough that Corollary 4.2 applies with $\eta = \eta_n$ for all sufficiently large n . Let G have e edges and r colors, with every C_7 rainbow, and let H be its cleaned subgraph. Put $q = e/n^2$ and $Q = e(H)/n^2$, so $0 \leq q - Q \leq \eta_n$.

Give every vertex of H weight $1/n$. Each edge has weight $1/n^2$, and each color class of H is independent in $J_{23}(H)$ by Corollary 4.2. Assigning weight $1/n^2$ to each nonempty such class gives $C \leq r/n^2$, where C is the cost in Theorem 3.2 for H with these weights. If $Q > 1/4$, Theorem 3.2 therefore yields

$$\frac{r}{n^2} \geq f(Q) \geq f(q) - \eta_n - \frac{1}{4} \sqrt{\eta_n},$$

where the last inequality uses $\sqrt{a} - \sqrt{b} \leq \sqrt{a - b}$ for $a \geq b \geq 0$.

If $Q \leq 1/4$, then $q \leq 1/4 + \eta_n$. Delete edges from G until $\lfloor n^2/4 \rfloor + 1$ remain. Theorem 1.1 gives

$$\frac{r}{n^2} \geq \frac{1}{8} - o(1) \geq f(q) - \eta_n - \frac{1}{4} \sqrt{\eta_n} - o(1).$$

Since $\eta_n \rightarrow 0$ and the error in Theorem 1.1 depends only on n , both estimates give (1) uniformly in e . $\square$

7 An upper bound at interior densities

We prove Theorem 1.3 with the chain of five cliques in Figure 1. Every edge lies inside one clique or joins two consecutive cliques, so a path needs at least $|i - j|$ edges to get from the i th clique of the chain to the j th. Each color is used at most twice, and the endpoints of two edges of the same color are far apart: however their endpoints are matched in pairs, the two pairs are in total at least six steps apart along the chain, so any cycle through both edges has at least $2 + 6 = 8$ edges.

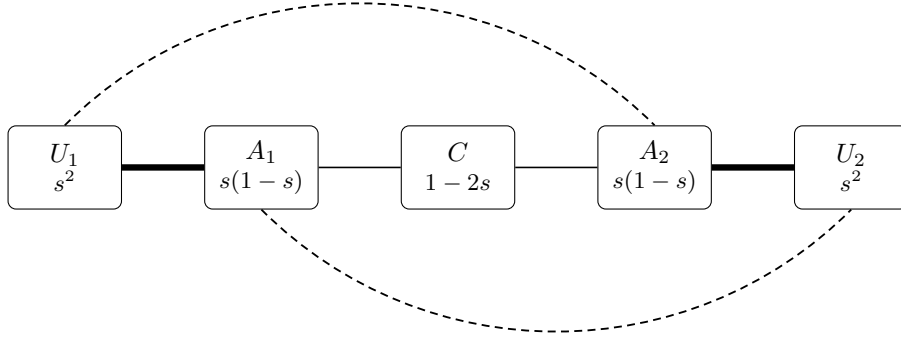


Figure 1: The five cliques in the proof of Theorem 1.3, with their vertex proportions; consecutive cliques are completely joined. The two thick joins are colored injectively with a common palette. The dashed arcs indicate that each edge inside U_1 shares its color with a distinct edge inside A_2 , and each edge inside U_2 with a distinct edge inside A_1 . All other edges receive distinct new colors.

Proof of Theorem 1.3. Since $\frac{1}{2} - 3s^2 + 4s^3$ decreases from $1/2$ to $1/4$ on $[0, 1/2]$, the parameter s is well defined. Take five disjoint cliques U_1, A_1, C, A_2, U_2 , in this order, with respective vertex proportions u, a, c, a, u , where

$$u = s^2, \quad a = s(1 - s), \quad c = 1 - 2s.$$

Join consecutive cliques completely and add no other edges between them. Color the two bipartite graphs U_1-A_1 and A_2-U_2 injectively with the same set of colors. Also pair each edge inside U_1 with a distinct edge inside A_2 , and each edge inside U_2 with a distinct edge inside A_1 ; these pairings are possible because $u \leq a$. Use a separate new color for each pair and for each remaining edge. Call the resulting graph G .

Every seven-cycle is rainbow. Indeed, if a cycle contains an edge inside U_1 and one inside A_2 , the two paths left after removing these edges each have length at least three. The same holds for edges inside U_2 and A_1 . For an edge of U_1-A_1 and an edge of A_2-U_2 , the corresponding paths have total length at least $3 + 3$ or $4 + 2$, according to the pairing of endpoints. In each case the cycle has at least eight edges, so it cannot be a seven-cycle.

The edge and color counts are

$$\begin{aligned} e(G) &= \left(\frac{c^2}{2} + a^2 + u^2 + 2ca + 2au \right) n^2 + O(n) = qn^2 + O(n), \\ r(G) &= e(G) - (au + u^2)n^2 + O(n) = (q - s^3)n^2 + O(n). \end{aligned}$$

Here the two bipartite graphs save $au n^2 + O(n)$ colors, and the two pairings of internal edges save $u^2 n^2 + O(n)$ colors. Symmetric rounding, which keeps $|U_1| = |U_2|$ and $|A_1| = |A_2|$, changes these counts by $O(n)$. Since $dq/ds = -6s(1-2s) < 0$, decreasing s by $O(1/n)$ if necessary and then deleting excess edges gives the stated upper bound at exactly $\lfloor qn^2 \rfloor$ edges.

This upper coefficient is strictly smaller than the one in [2]. Indeed,

$$q - \frac{1}{4} - (q - 2s^3)^2 = s^3(1 - 2s)^2(2 - s) > 0,$$

and hence

$$q - s^3 < \frac{q}{2} + \frac{1}{2} \sqrt{q - \frac{1}{4}}.$$

□

8 Further questions

1. Is Theorem 1.3 sharp? That is, for fixed $1/4 < q < 1/2$ and s as in the theorem, is

$$\chi_S(n, \lfloor qn^2 \rfloor, C_7) = (q - s^3)n^2 + o(n^2)?$$

The lower bound of Theorem 1.2 is strictly smaller at every fixed interior density.

2. If an n -vertex graph G_n has $e(G_n) = \lfloor n^2/4 \rfloor + 1$ and admits a coloring with $n^2/8 + o(n^2)$ colors in which every C_7 is rainbow, must G_n differ from the disjoint union of two cliques of orders $n/2 + o(n)$ by only $o(n^2)$ edge additions and deletions?

The argument of Section 6 gives a partial answer: the normalized degree variance of such graphs tends to zero. Indeed, if $\sigma_n^2 \geq v > 0$, running that argument with cleaning parameter $\eta = v^2/128$ and reweighting parameter $\lambda = v/16$ in place of γ gives $Q(w) \geq 1/4 + v^2/64$. Theorem 3.2 then forces more than $(1/8 + v/70)n^2$ colors for large n .

3. Does every n -vertex graph with more than $\lfloor n^2/4 \rfloor$ edges contain a family of at least $n^2/8 - o(n^2)$ edges such that every two lie on a common C_7 ?

A positive answer would strengthen Theorem 1.1, since all edges in such a family must receive different colors. Section 5 constructs such families under its nearly regular hypotheses. The fractional coloring argument in Section 3 does not produce such a family in general.

4. Can Theorem 1.1 be sharpened to

$$\chi_S(n, \lfloor n^2/4 \rfloor + 1, C_7) = \frac{n^2}{8} + O(n)?$$

The upper bound already has this error term.

5. Can Theorem 1.1 be proved without Szemerédi’s regularity lemma? In particular, can a direct structural argument replace the passage from closed walks to cycles in Section 4?

Such an argument could improve the quantitative bounds. Bucić, Chen and Ma [2, Sections 2–4] prove their result for longer odd cycles using induction and short-path arguments.

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